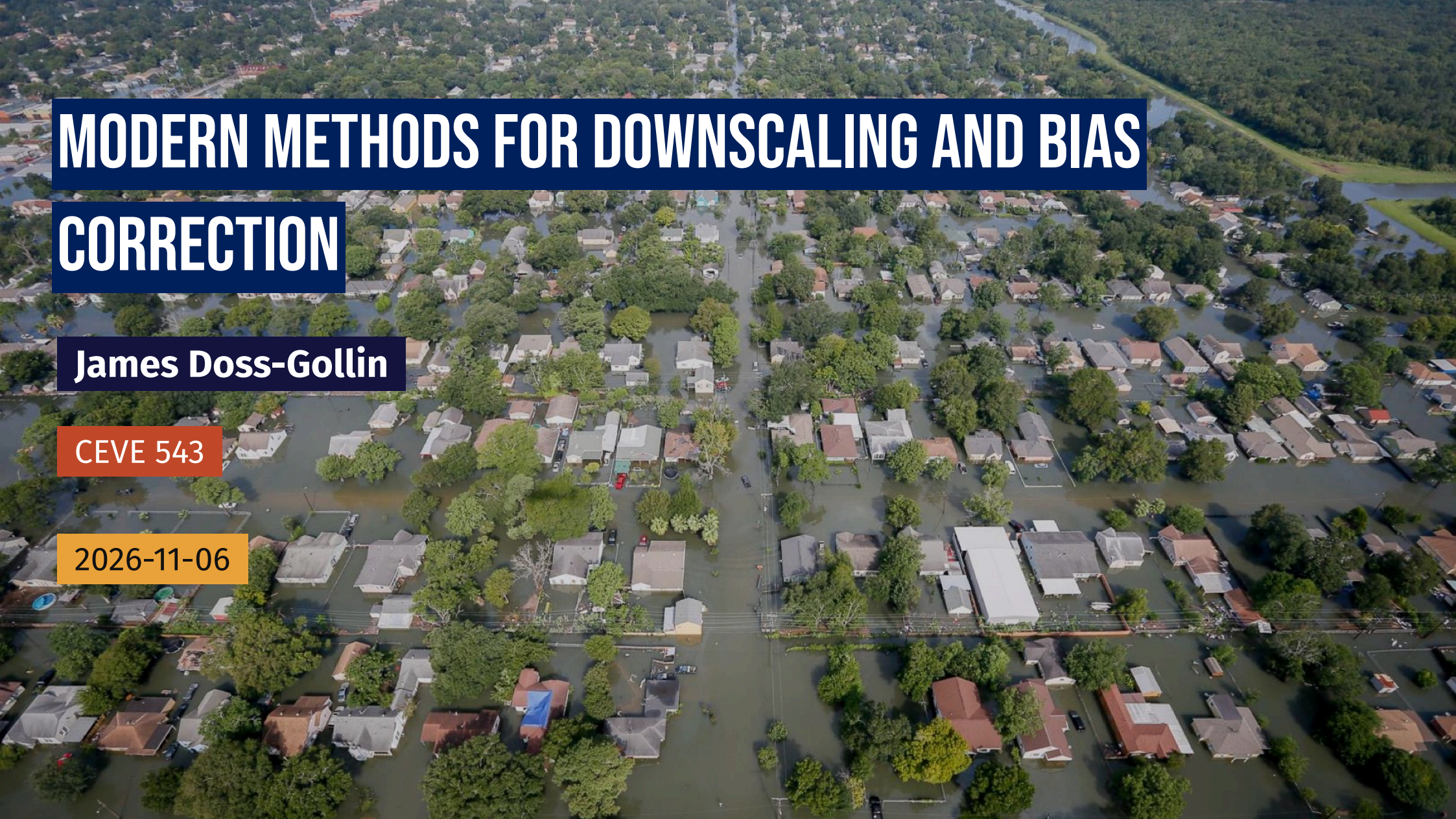
# MODERN METHODS FOR DOWNSCALING AND BIAS CORRECTION

James Doss-Gollin

CEVE 543

2026-11-06



**DeepSD**

CorrDiff

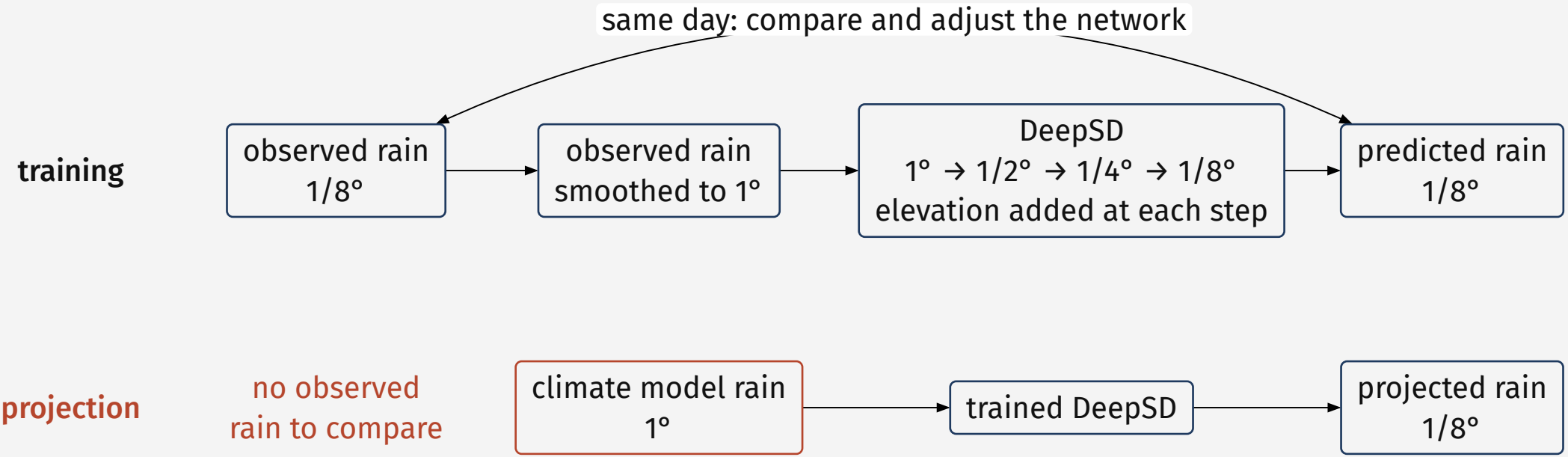
# DOWNSCALING AS IMAGE SUPER-RESOLUTION

**i Note**

Figure shown in class: Prism Observed Precipitation: A) Low resolution at 1.0◦ (~ 100km). B) High resolution at 1/8◦ (~ 12.5km). Vandal et al. (2017), Fig. 1.

- Training:
  - PRISM, United States, 1980-2005.
  - Target: observed rain at 1/8°
  - Predictor: same data coarsened to 1°
- Projection: a climate model's 1° rain, which has no pair
- Ill-posed problem (many possible solutions)

# HOW DEEPSD IS TRAINED AND USED



DeepSD

# CorrDiff

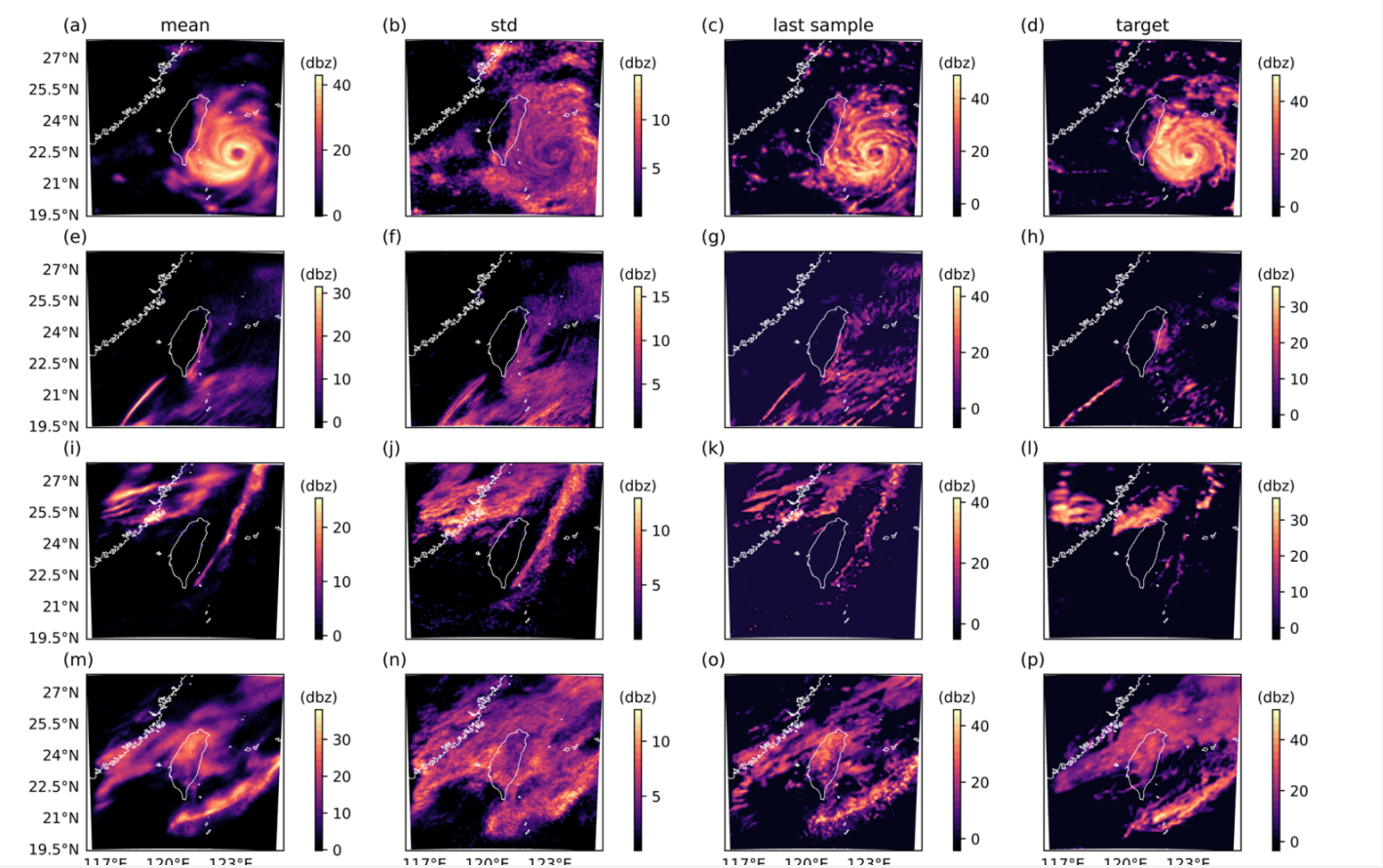
# CorrDiff predicts a mean field, then generates detail

**Note**

Figure shown in class: Regression and diffusion steps from 25 km to 2 km. Mardani et al. (2025), Fig. 6.

- A network predicts the mean 2 km field; a diffusion model samples the detail around it.
- Trained on 3 years over Taiwan, from 25 km reanalysis to a 2 km weather model.
- In: 12 reanalysis variables. Out: 2 m temperature, 10 m winds, and radar reflectivity.

# GENERATED RADAR FOR A TYPHOON, A FRONT, AND TWO RANDOM DATES

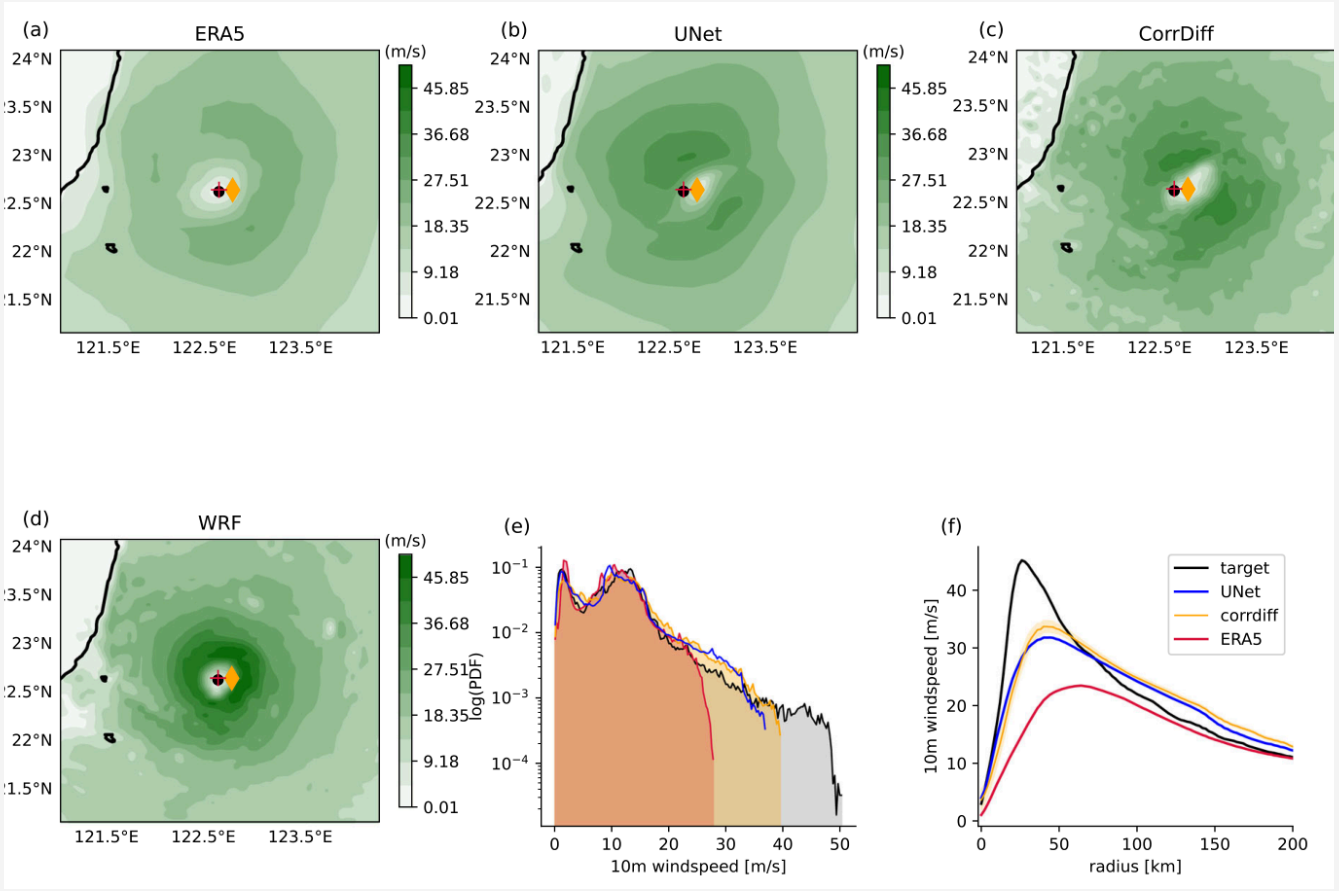



Ensemble mean, spread, one sample, and the target, four dates. Mardani et al. (2025), Fig. 3 (CC BY-NC-ND 4.0).

Rows, top to bottom (UTC):

- Typhoon Haikui, 3 Sep 2023, 00:00
- Random date, 17 Feb 2021, 21:00
- Random date, 4 Mar 2021, 01:00
- Cold front, 13 Feb 2022, 20:00

# DOES CORRDIFF RECOVER TYPHOON HAIKUI'S WINDS?




Haikui's 10 m winds from reanalysis, two models, and the target. Mardani et al. (2025), Fig. 5 (CC BY-NC-ND 4.0).

Peak wind around the center:

- reanalysis 22 m/s
- CorrDiff 33 m/s
- 2 km model 45 m/s